

SWARAJ LADKE

Applied AI & Full Stack Engineer

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CAREER SUMMARY

Applied AI and full-stack engineer who ships machine learning systems end to end — from CPU-bound real-time inference to the production interfaces around it. Recent work includes a real-time neuromorphic vision detector submitted to the TII OrbitSight Challenge (Technology Innovation Institute, Abu Dhabi) and independent research on continual learning. Oracle-certified in Generative AI and Cloud, with peer-reviewed publications.

TECHNICAL SKILLS

- **Languages:** Python, JavaScript, TypeScript.
- **AI / ML:** PyTorch, TensorFlow, Scikit-learn, OpenCV, Event-Based Vision, Hugging Face Transformers, CUDA, Continual Learning, Predictive Coding, LangChain, NLP, LLMs, RAG, LoRA, AI Agents.
- **Frontend:** React, Next.js, Tailwind CSS, ShadCN, Fabric.js, Redux, Framer Motion, Three.js.
- **Backend & Databases:** FastAPI, Flask, Node.js, MongoDB, PostgreSQL, MySQL, Supabase, ChromaDB, WebSockets, REST APIs.
- **Cloud & DevOps:** AWS, Azure, Google Cloud, Vertex AI, Docker, Vercel, Render, GitHub Actions.
- **Developer Tools:** Postman, Vite, LM Studio, Hugging Face Spaces, Tauri, OAuth, Google Colab, Kaggle, MLflow.

WORK EXPERIENCE

Applied AI Engineering Intern | OVAA Technologies Dec 2025 – May 2026

- Engineered AI-powered applications using Large Language Models, Retrieval-Augmented Generation, and AI APIs to deliver intelligent software solutions.
- Developed scalable backend services in Python and Flask, integrated AI workflows, and contributed to the end-to-end development, testing, and deployment of production-ready AI applications.

Full Stack Developer Intern | TechnoHacks Solutions Pvt. Ltd. Jun 2025 – Jul 2025

- Built and deployed scalable full-stack web applications using React.js, Next.js, TypeScript, Flask, MongoDB, and REST APIs.

RELEVANT PROJECTS

OrbitSight — Real-Time Space-Object Detection from Neuromorphic Event Cameras TII Challenge, 2026

- Built an event-native detection pipeline for resident space objects from raw neuromorphic sensor streams, submitted to the TII OrbitSight Challenge (Technology Innovation Institute, Abu Dhabi); scored **0.284 mAP@0.5** across 21 sequences, verified against the organiser's evaluator.
- Designed a post-hoc box regressor that improved mAP by **49.1%** over the geometric baseline (+125.2% on sparse sequences), recovering 61.1% of the measured oracle headroom.
- Benchmarked three candidate architectures (classical CV, CNN, spiking neural network) and shipped the lightest on measured evidence — 47K parameters, **4.9–31.3 ms p99 latency** against a 40 ms budget, CPU-only and fully offline in Docker.
- **Tech Stack:** Python, NumPy, Scikit-learn, OpenCV, Docker, Event-Based Vision.

AGNIS — Continual Learning Architecture & Audited Benchmark Suite Ongoing Research

- Rebuilt an exploratory continual-learning project around a reproducible class-incremental benchmark (Split-CIFAR-100, ResNet-18, 10 sequential tasks, 5 seeds) with pre-registered halt criteria, retracting earlier unaudited results.
- Developed an audit toolchain enforcing 21 standing methodology rules — verifying every reported number against committed run logs and failing the build on violation.
- **Tech Stack:** Python, PyTorch, CUDA, Scikit-learn, Predictive Coding.

HireFlow AI — Autonomous AI Recruitment & Proctoring Platform

- Engineered an end-to-end, multi-service recruitment platform with an ATS resume screener, an AI-driven behavioural interviewer, and a locally-proctored live coding environment for autonomous candidate evaluation.
- **Tech Stack:** React, Next.js, Python, Flask, SQLite, OpenCV, REST APIs, Micro-frontends.

Additional live projects: AGNIS Drive (cloud drive, real-time sync — React 19, Next.js 15, Appwrite), AGNIS Design Studio (collaborative design platform — Next.js, Liveblocks, Fabric.js), Recasta Studios (AI content repurposing — React, Node.js).

EDUCATION

Master of Computer Applications | Savitribai Phule Pune University 2024 – 2026

Bachelor of Computer Applications | Savitribai Phule Pune University 2021 – 2024

CERTIFICATIONS & PUBLICATIONS

- **Oracle Cloud Infrastructure 2025:** Certified Generative AI Professional; AI Foundations Associate; APEX Cloud Certified Developer Professional.
- **Short courses:** AI & Career Empowerment — R. H. Smith School, Univ. of Maryland, USA; AI for Work & Life — Univ. of North Florida, USA.
- Emoify (Research Paper): zenodo.org/records/15752110
- AI Augmented DigitalTwin (Research Paper): ijsci.com/index.php/home/article/view/744